

VLAN Trunking and Access Ports Theory

Task 1 :- In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol.

ANS :-

1. **Connect Switches** : Drag Switch1 and Switch2 to the workspace and connect them from Fa0/1 to Fa0/1 using a crossover cable.
2. **Configure Switch1** : Click Switch1 → CLI tab → press Enter:

```
enable
configure terminal
interface fa0/1
switchport mode trunk
```

3. **Configure Switch2** : Click Switch2 → CLI tab → press Enter:

```
enable
configure terminal
interface fa0/1
switchport mode trunk
```

Task 2 :- Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

ANS :-

- 1. On Both Switches (Switch1 & Switch2): Open CLI and run :**

```
enable
configure terminal
vlan 10
vlan 20
vlan 30
```

- 2. On Switch1 Only: Assign ports to VLANs :**

```
interface fa0/2
switchport mode access
switchport access vlan 10

interface fa0/3
switchport mode access
switchport access vlan 20

interface fa0/4
switchport mode access
switchport access vlan 30
end
```

Task 3 :- Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN.

ANS :-

1. Connect & IP PCs on Switch1 :-

- PC1 → Fa0/2 (VLAN 10) | IP: 192.168.10.1
- PC2 → Fa0/3 (VLAN 20) | IP: 192.168.20.1
- PC3 → Fa0/4 (VLAN 30) | IP: 192.168.30.1

2. Test Communication :-

- From PC1, ping PC2 (192.168.20.1): FAILS (Different VLANs are isolated).
- From PC1, ping another PC in VLAN 10: SUCCEEDS (Same VLANs can communicate).

Task 4 :- Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches.

ANS :-

1. Change Encapsulation (if supported by switch model) :

```
enable
configure terminal
interface fa0/1
switchport trunk encapsulation isl
```

2. Test Communication:

Ping between PCs in the same VLAN across switches.

3. Result :

If both ends match (ISL ↔ ISL): VLAN traffic passes.

If standard 2960 switches are used: The command is rejected, so the link stays 802.1Q and traffic continues to pass.

If there is an encapsulation mismatch (ISL on one end, 802.1Q on the other): The trunk drops and VLAN traffic fails.

Task 5 :- Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.

ANS :-

1. Configure Trunk Mode on Fa0/5 :

```
enable
configure terminal
interface fa0/5
switchport mode trunk
```

2. Connect PC to Fa0/5 (assigned IP in VLAN 20 range) :

Try pinging another PC in VLAN 20.

3. Result / Impact :

- Ping FAILS : PCs send standard untagged frames and cannot read tagged VLAN frames.
- A trunk port expects tagged frames (for non-native VLANs like VLAN 20). Since the PC sends untagged traffic, the switch sends it to the default Native VLAN (VLAN 1) instead of VLAN 20, cutting off communication with VLAN 20 devices.